<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks Available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>1</td><td>(a)</td><td></td><td></td><td>Windows at 15% — 4 correct 3 marks
Floor at 10% — 2 or 3 correct 2 marks
Roof at 25% — 1 correct 1 mark
Walls at 35% — 0 correct 0 mark</td><td></td><td>3</td><td></td><td>3</td><td>1</td><td></td></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td></td><td>Fibre-glass is an insulator or it contains air or it contains air pockets or <u>reduces</u> conduction through the ceiling (1) Don't accept traps air
<u>Reduces</u> convection <u>currents</u> in the loft or air in the loft heats up less (1)</td><td>2</td><td></td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td></td><td>3 × (1): Ticks in boxes alongside statements 2, 4 and 5 i.e.
The required thickness of loft insulation in 2000 is 8 times thicker than in 1970.
A house built in 1980 needs 210 mm of loft insulation added to bring it up to 2015 standards.
The general trend of the graph indicates that the thickness of required loft insulation has increased at an increasing rate.
Deduct 1 mark for each additional tick</td><td></td><td>3</td><td></td><td>3</td><td>3</td><td></td></tr>
<tr><td></td><td></td><td>(iii)</td><td></td><td>$4.50 \times 120 = £540$ (1)
$\frac{540}{98} = 5.51$ or 5.5 or 6 [years] (1)
Don't accept 5 [years]
Answer of 0.0459 [years] award 1 mark only</td><td></td><td>2</td><td></td><td>2</td><td>2</td><td></td></tr>
<tr><td></td><td></td><td>(iv)</td><td></td><td>(1) for either calculation:
Insulation 1 saving $(40 - 5.0) \times £84 = £2\,940$
Insulation 2 saving $(40 - 6.0) \times £111 = £3\,774$
(1) for second calculation and comment/agreement with builder

Alternative 1:
(1) for either calculation:
$5 \times 84 = £420$ and $40 \times 84 = £3\,360$ and the difference = £2 940
$6 \times 111 = £666$ and $40 \times 111 = £4\,440$ and the difference = £3 774
(1) for second calculation and comment/agreement with builder</td><td></td><td></td><td>2</td><td>2</td><td>1</td><td></td></tr>
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Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks Available					
				Alternative 2: 5 × 84 = £420 **and** 6 × 111 = £666 **or** 40 × 84 = £3 360 **and** 40 × 111 = £4 440 (1) The differences are £3 774 **and** £2 940 so the builder is correct (1) **Alternative 3:** **(1) for either calculation:** 3.5 × 120 = £420 **and** 40 × 84 = £3 360 **and** the difference = £2 940 5.5 × 120 = £666 **and** 40 × 111 = £4 440 **and** the difference = £3 774 **(1) for second calculation and comment/agreement with builder** **Alternative 4:** 3.5 × 120 = £420 **and** 5.5 × 120 = £666 **or** 40 × 84 = £3 360 **and** 40 × 111 = £4 440 (1) The differences are £3 774 **and** £2 940 so the builder is correct (1) **Alternative 5:** **(1) for either calculation:** (111 – 84) × 40 = £1 080 **or** 666 – 420 = £246 £1 080 – £246 = £834 **(1) and comment/agreement with builder** N.B. Any reference to insulation 3 treat as neutral						
				Question 1 total	**2**	**8**	**2**	**12**	**7**	**0**